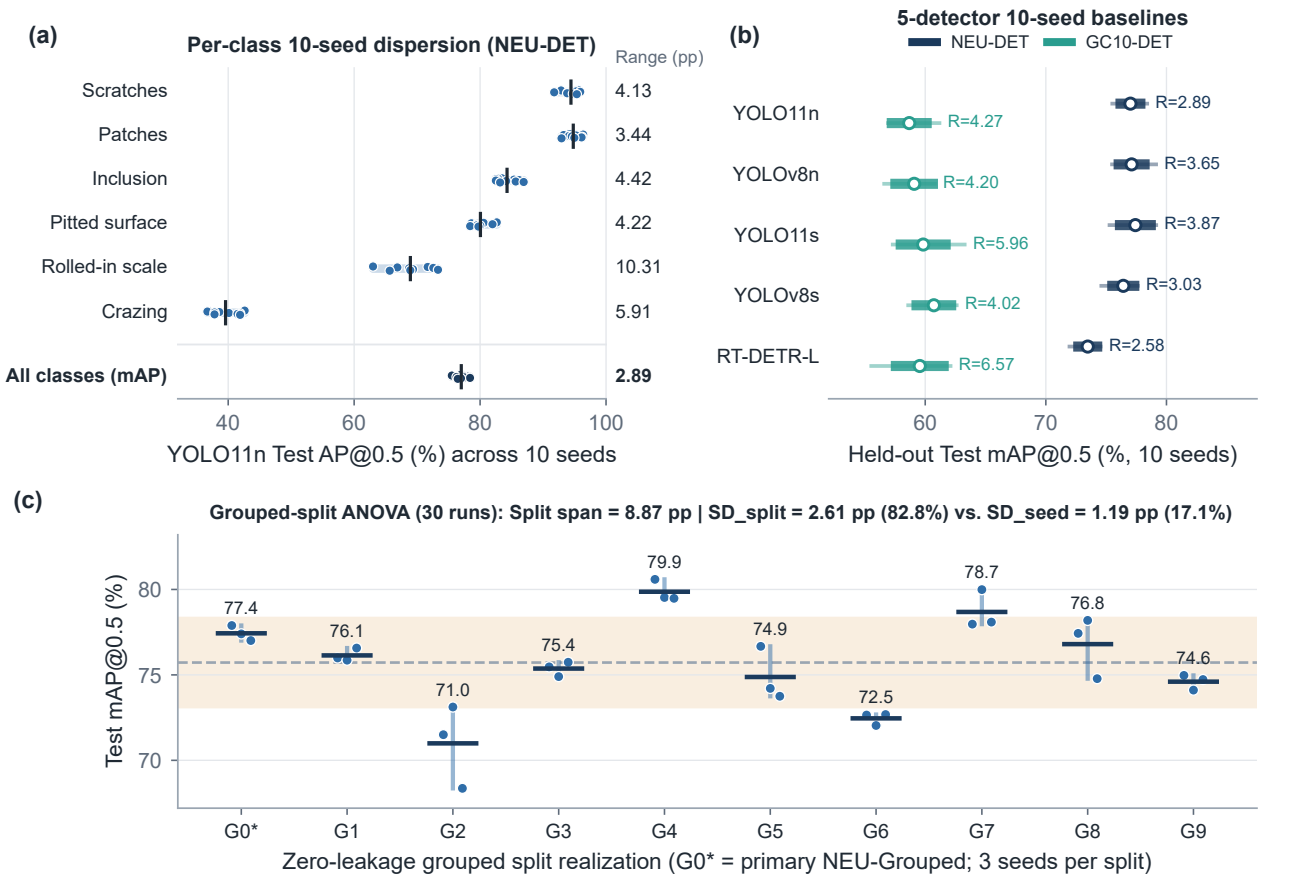

(a)
Per-class 10-seed dispersion (NEU-DET)
Range (pp)
Scratches
4.13
Patches
3.44
Inclusion
4.42
Pitted surface
4.22
Rolled-in scale
10.31
Crazing
5.91
All classes (mAP)
2.89
40
60
80
100
YOLO11n Test AP@0.5 (%) across 10 seeds
(b)
5-detector 10-seed baselines
NEU-DET
GC10-DET
YOLO11n
R=4.27
R=2.89
YOLOv8n
R=4.20
R=3.65
YOLO11s
R=5.96
R=3.87
YOLOv8s
R=4.02
R=3.03
RT-DETR-L
R=6.57
R=2.58
60
70
80
Held-out Test mAP@0.5 (%, 10 seeds)
(c)
Grouped-split ANOVA (30 runs): Split span = 8.87 pp | SD_split = 2.61 pp (82.8%) vs. SD_seed = 1.19 pp (17.1%)
Test mAP@0.5 (%)
80
75
70
77.4
76.1
71.0
75.4
79.9
74.9
72.5
78.7
76.8
74.6
G0*
G1
G2
G3
G4
G5
G6
G7
G8
G9
Zero-leakage grouped split realization (G0* = primary NEU-Grouped; 3 seeds per split)